Julien Audiffren, Ph.D.

julien.audiffren@gmail.com
https://scholar.google.fr/citations?user=7jcEx7cAAAAJ&hl=en
https://jythen.github.io/
https://www.linkedin.com/in/julien-audiffren/

Employment History

2018 – now — **Senior Researcher.** Exascale Infolab & Copelab, University of Fribourg, Fribourg, Switzerland

2014 – 2018 — **Research Associate.** Centre de Mathematique et Leurs Applications (CMLA), ENS Paris Saclay, Paris, France

2012 – 2014 — **Posdoctoral Researcher.** Laboratoire d'Informatique Fondamentale (LIF), Aix Marseille University, Marseille, France

Education

2008 – 2011 — **Ph.D., ENS Cachan & Aix Marseille University** Department of Mathematics
Thesis title: *Muller's ratchet: Asymptotic Behavior of Stochastic Differential Equations.*

2004–2007 — **Master & Aggregation, Ecole Normale Superieure de Cachan**.
Mathematics, Probability and Statistics.

Skills

Languages — French (Native); English (Proficient); German(Intermediate)

Coding — Python, C #, LaTeX

Research Interests — Reinforcement Learning, Multi-armed Bandits, Global Optimization, Human assisting AI, Machine learning for Healthcare

Teaching. — Advanced Probability and statistics, Linear Algebra, Machine Learning, Algorithmic, Optimization, Behavioral Game Theory, Psychophysics

Miscellaneous Experience

PhD (Co)Supervision

- **Humbert, P.** State of PhD : Defended January, 2021.
- **Nicolaï, A.** State of PhD : 3nd Year
- **Rettig, L.** State of PhD : 4th Year

Scientific Popularization

- Co-founder of 13 Minutes Marseille https://treize.lis-lab.fr/
- Writer and Contributor to the Data Analytic Post https://dataanalyticspost.com/
- Co-organizer of the AI Challenge For Industry 2019 https://www.aichallenge.parisregion.eu/industry-challenge/

References

Available on Request

Selected Research Publications

Complete list available on Google Scholar

Journal Articles

1. Humbert, P., Dubost, C., Audiffren, J., & Oudre, L. (2019). Apprenticeship Learning for a Predictive State Representation of Anesthesia. *IEEE Transactions on Biomedical Engineering*.
2. Minvielle, L., & Audiffren, J. (2019). NurseNet: Monitoring Elderly Levels of Activity with a Piezoelectric Floor. *Sensors*, *19*(18), 3851.
3. Bargiotas, I., Audiffren, J., Vayatis, N., Vidal, P.-P., Buffat, S., Yelnik, A. P., & Ricard, D. (2018). On the importance of local dynamics in statokinesigram: A multivariate approach for postural control evaluation in elderly. *PloS one*, *13*(2).

Conference Proceedings

1. Audiffren, J. (2021). Multi-armed bandits for quantifying human perception with dichotomous optimistic search. In *20th international conference on autonomous agents and multiagent systems*.
2. Nicolaï, A., & Audiffren, J. (2018). Model-space regularization and fully interpretable algorithms for postural control quantification. In *2018 IEEE 42nd Annual Computer Software and Applications Conference (COMPSAC)* (Vol. 2, pp. 177–182). IEEE.
3. Audiffren, J., & Ralaivola, L. (2017). Bandits dueling on partially ordered sets. In *Advances in Neural Information Processing Systems* (pp. 2129–2138).

Patents

1. Vayatis, N., Vidal, P., Audiffren, J., Yelnik, A., Ricard, D., Bargiotas, I., … Buffat, S. (2020). *Improved method for quantifying balance*. US Patent App. 16/763,573.
2. Vayatis, N., Vidal, P., Promponas, N., Audiffren, J., Yelnik, A., De Waele, C., & Ricard, D. (2020). *Multiparameter method for quantifying balance*. US Patent App. 16/346,395.